What test to run?

Understanding Decision Split Criteria

Non-Parametric =
No parameters are estimated
No estimation (effect size or confidence interval), just hypothesis testing (p-value)

Independent observations?

Independence: Examples & Implications
What are some examples of correlated data?

- Siblings/Twins
- Same patients pre/post treatment
- Different body parts on the same person
- Multiple measures over time of the same person
- Cluster within the same school, teacher, physician

What happens if you ignore correlations?

- Ignoring between person/group: Underestimate p-values (p-values are too small).
  - Incorrectly decrease your Type I error rate. More likely to declare a significant test result when there isn't one.
  - Artificially inflate the sample size --> pool together samples when they should be separate
- Ignoring within-person/group: overestimate p-values (p-values are too high)
  - Variance is artificially high: less variability between subjects is less than for independent samples

[see source: Stanford Medical Stats]

Normal outcomes?

Parametric Assumption #1
Normality:

- Visual: Plot a histogram, normal probability plot
- Stats: descriptive stats such as mean aprox = median, 2/3 observations fall within 1 std dev of mean
- Tests: Kolmogorov-Smirnov (influenced by ss), Shapiro-Wilk test

For Linear Models:

- Visual: **Residual** plots are normally distributed, use a Quantile plot (QQ plot)

Scenarios
1.Non-Normal Dependent Variable

- IF SS isn't small, the ***means*** of the outcome variable will be normally distributed thanks to the **Central Limit** even if the underlying variable is not (use bootstrap to re-sample)
- Transform dependent variable OR use non-parametric tests when:
  - SS is small AND non-normal outcome
  - SS is unequal AND variances are non-homogenous

[see source: Stanford Medical Stats]

Equal outcome variances?

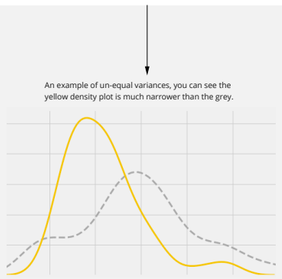
Parametric Assumption #2
How to validate parametric assumptions?
Homogeneous Variances:

- Visual: plot histograms
- Stats: values of variances/standard deviation
- Tests: Lavene, Equality of variances F-test (low p-value mean variances are unequal)

For Linear Models:

- Visual: Scatter around the linear fit
- Tests: Brown-Forsythe for Anova test for equal variances

[see source: Stanford Medical Stats]



Commonly Used in Online Site Testing